

# Contact Management System

## Project Overview

I created this contact management system to get better at Python! It works like a phonebook on your computer - you type in names and phone numbers, and it remembers them. When you close the program and open it again, all your contacts are still there because they get saved to a file. Pretty cool for a beginner project!

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## What I Learned Building This

### 1. File Handling

- How to open files and read what's inside them
- Saving information to files so it doesn't disappear
- The with open() trick that closes files automatically (super helpful!)
- What to do when a file doesn't exist yet (try-except blocks)
- Making data stick around between program runs

### 2. Data Structures

- Dictionaries are awesome! They pair names with phone numbers perfectly
- How to check if something exists in a dictionary
- Different ways to loop through dictionaries
- Why dictionaries are better than lists for this project

### 3. String Manipulation

- Breaking up text lines with .split(',')
- Getting rid of extra spaces with .strip()
- Putting strings together with f-strings
- Writing data in CSV style (comma-separated values)

### 4. Functions & Organization

- Breaking my code into smaller pieces (functions)
- Making functions do one specific job
- How to pass information into functions
- Using a global dictionary that all functions can access

### 5. Program Flow Control

- Creating a menu that keeps running until you want to quit
- Using numbers to pick different actions
- While loops that go forever (until you break out)

- If-elif-else for handling menu choices

## 6. Error Handling

- Stopping the program from crashing when files are missing
- Showing helpful messages instead of scary error codes
- Try-except blocks saved my program multiple times!
- Starting fresh when there's no data file

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### How to Run

# Step 1: Check if Python is installed

```
python --version
```

# Step 2: Go to your project folder

```
cd path/to/project
```

# Step 3: Start the program

```
python contact_manager.py
```

# Step 4: Use the menu to add/search contacts!

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### Features Breakdown

| Feature | What It Does |
|---------|--------------|
|---------|--------------|

|                    |                                         |
|--------------------|-----------------------------------------|
| <b>Add Contact</b> | Type a name and phone number to save it |
|--------------------|-----------------------------------------|

|                       |                                               |
|-----------------------|-----------------------------------------------|
| <b>Search Contact</b> | Look up someone's number by typing their name |
|-----------------------|-----------------------------------------------|

|                    |                                      |
|--------------------|--------------------------------------|
| <b>Display All</b> | See your entire contact list at once |
|--------------------|--------------------------------------|

|                  |                                                |
|------------------|------------------------------------------------|
| <b>Auto-Save</b> | Everything saves automatically to contacts.txt |
|------------------|------------------------------------------------|

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| <b>Load on Start</b> | Your contacts appear every time you run the program |
|----------------------|-----------------------------------------------------|

|                    |                                                 |
|--------------------|-------------------------------------------------|
| <b>Simple Menu</b> | Easy numbered options - just type 1, 2, 3, or 4 |
|--------------------|-------------------------------------------------|

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### How Data Gets Stored

The program creates a file called contacts.txt that looks like this:

```
John Doe,555-1234
```

Jane Smith,555-5678

Bob Wilson,555-9012

Each line has a name, a comma, and a phone number. Simple but effective!

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### **Sample Session**

Contacts loaded successfully.

--- Contact Management System ---

1. Add Contact
2. Search Contact
3. Display All Contacts
4. Exit

Enter your choice: 1

Enter name: Mike Johnson

Enter phone number: 555-7890

Contacts saved successfully.

Contact 'Mike Johnson' added successfully.

--- Contact Management System ---

1. Add Contact
2. Search Contact
3. Display All Contacts
4. Exit

Enter your choice: 2

Enter name to search: Mike Johnson

Name: Mike Johnson, Phone: 555-7890

--- Contact Management System ---

1. Add Contact
2. Search Contact
3. Display All Contacts
4. Exit

Enter your choice: 3

All Contacts:

Name: Mike Johnson, Phone: 555-7890

--- Contact Management System ---

1. Add Contact
2. Search Contact
3. Display All Contacts
4. Exit

Enter your choice: 4

Exiting program.

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## Challenges I Faced & How I Solved Them

### Challenge 1: My contacts kept disappearing when I closed the program!

- **Solution:** I learned about file handling! Now everything gets written to contacts.txt and loaded back when the program starts.

### Challenge 2: The program crashed when contacts.txt didn't exist

- **Solution:** Wrapped the file reading in a try-except block. Now it just prints a message and starts with an empty contact list.

### Challenge 3: How do I store multiple contacts efficiently?

- **Solution:** Dictionaries! They're perfect because I can use the name as a key to quickly find phone numbers.

### Challenge 4: Reading the file gave me messy data with extra spaces

- **Solution:** Used .strip() to remove whitespace and .split(',') to separate names from phone numbers cleanly.

### Challenge 5: Needed to save contacts every time I added one

- **Solution:** Called save\_contacts() inside the add\_contact() function. Automatic saving!

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## What I'd Improve Next

- **Update Contacts:** Let users change phone numbers for existing contacts
- **Delete Contacts:** Remove contacts you don't need anymore
- **Better Validation:** Check if phone numbers are actually numbers
- **Multiple Numbers:** Store work, home, and mobile numbers per person

- **Email Addresses:** Add email field for each contact
  - **Categories:** Tag contacts as family, friends, work, etc.
  - **Backup Feature:** Export contacts to a backup file
  - **Import Feature:** Load contacts from different files
  - **Better File Format:** Switch to JSON or CSV for more complex data
  - **Search Improvements:** Find contacts even if you misspell the name
  - **GUI Version:** Build a graphical interface with Tkinter
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## Code Structure

### Global Variable

- `contacts = {}` - Dictionary storing all contact information

### Function: `load_contacts()`

**What it does:** Reads `contacts.txt` and fills the contacts dictionary

**How it works:**

1. Opens `contacts.txt` in read mode
2. Reads each line one by one
3. Splits each line at the comma
4. Puts name and phone into the dictionary
5. Handles the error if file doesn't exist

### Function: `save_contacts()`

**What it does:** Writes all contacts from dictionary to `contacts.txt`

**How it works:**

1. Opens `contacts.txt` in write mode (creates if missing)
2. Loops through each name and phone in dictionary
3. Writes each contact as `"name,phone\n"`
4. Confirms save was successful

### Function: `add_contact(name, phone)`

**Parameters:**

- `name` (string) - Person's name
- `phone` (string) - Phone number

**What it does:** Adds a new contact and saves it immediately

**How it works:**

1. Puts the name and phone into contacts dictionary
2. Calls `save_contacts()` to write to file
3. Prints success message

**Function: `search_contact(name)`**

**Parameters:**

- `name` (string) - Name to search for

**What it does:** Looks up a contact and displays their info

**How it works:**

1. Checks if name exists in dictionary
2. If found, prints name and phone
3. If not found, prints "not found" message

**Function: `display_contacts()`**

**What it does:** Shows all contacts in a list

**How it works:**

1. Checks if contacts dictionary has anything
2. If yes, loops through and prints each contact
3. If empty, tells user no contacts exist

**Function: `main()`**

**What it does:** Runs the whole program

**How it works:**

1. Loads contacts from file at startup
2. Shows menu in infinite loop
3. Gets user's choice
4. Calls the right function based on choice
5. Exits when user picks option 4

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**Files Created**

- `contact_manager.py` - Main program with all the code
- `contacts.txt` - Automatically created to store your contacts

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**Key Python Concepts I Used**

Dictionaries for key-value storage  
File I/O (reading and writing)

Exception handling with try-except  
Functions with parameters  
While loops for menus  
If-elif-else conditionals  
String methods (split, strip)  
F-strings for formatting  
Global variables  
Context managers (with statement)

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## **Learning Outcomes**

After finishing this project, I got way better at:

- Working with files in Python
- Organizing data with dictionaries
- Handling errors without crashing
- Breaking programs into logical functions
- Creating user-friendly menus
- Making data persist between sessions
- Reading and writing text files
- Parsing CSV-style data

This was honestly one of my favorite beginner projects because I could actually use it in real life!

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## **Tips for Beginners**

If you're building something similar, here's what helped me:

1. Start with the dictionary structure first before adding files
2. Test each function individually before connecting everything
3. Always use with open() - it's safer than plain open()
4. Print statements everywhere while debugging!
5. Handle the "file not found" error early or you'll get frustrated
6. Save after every change so you don't lose data